

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Valve camshaft installation and removal service tool, Part Number 5299837, or equivalent
- Lubriplate™, Part Number 3163086, or equivalent
- Camshaft Installation and Removal Kit, Part Number 5395348, or equivalent.

Additional Service Items

- ENVIROSOL™ 655 solvent or equivalent
- A new, clean, lint-free cloth.

Preparatory Steps

WARNING

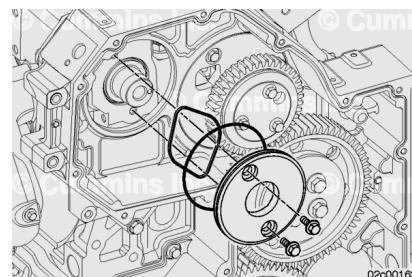
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the battery. See equipment manufacturer service information.
- Remove the exhaust gas recirculation (EGR) crossover tube. Refer to Procedure 011-070 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-070-tr.html>)
- Remove air compressor inlet tube, if equipped. Refer to Procedure 012-019 in Section 12. (</qs3/pubsys2/xml/en/procedures/377/377-012-019.html>)
- Remove the upper gear cover. Refer to Procedure 001-079 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-079.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/377/377-003-011-tr.html>)
- Disconnect the internal actuator wiring harness. Refer to Procedure 019-063 in Section 19. (</qs3/pubsys2/xml/en/procedures/377/377-019-063.html>)
- Remove the rocker lever shafts. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/377/377-003-009.html>)
- Remove the tone wheel and valve camshaft gear. Refer to Procedure 001-088 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-088.html>)

Remove

Remove the retainer capscrews, retainer, seal, and o-ring.

Discard the seal and o-ring.

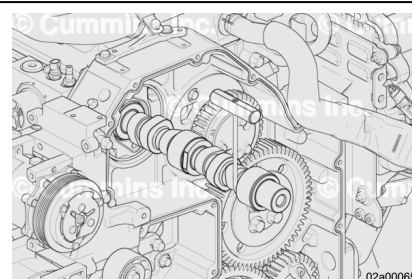


⚠ WARNING ⚠

To reduce the possibility of personal injury, wear protective gloves when handling parts with sharp edges.

⚠ CAUTION ⚠

Use caution when removing the camshaft to avoid damaging the camshaft bushings.



Use the Valve Camshaft Removal and Installation Kit, Part Number 5299837, to remove the camshaft from the cylinder head.

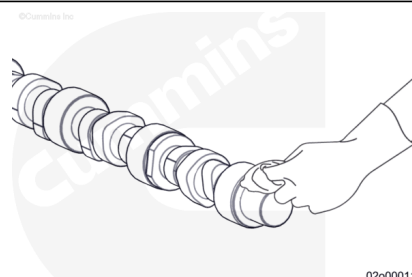
Use the Camshaft Removal and Installation Kit, Part Number 5395348, to pull and guide the camshaft out of the cylinder head until three journals remain in the cylinder head. Carefully remove the remainder of the camshaft by hand.

Clean and Inspect for Reuse

⚠ WARNING ⚠

To reduce the possibility of personal injury, wear protective gloves when handling parts with sharp edges.

⚠ WARNING ⚠



When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the valve camshaft with ENVIROSOL™ 655 solvent or equivalent, and a new, clean, lint-free cloth.

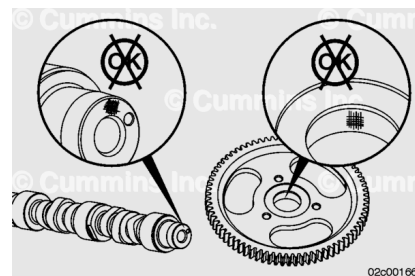
Do **not** spray solvent directly onto the camshaft nose. Spray solvent onto a new, clean, lint-free cloth. Wipe the camshaft nose clean. Do **not** spray the solvent directly onto the components. Make sure no oil residue remains after cleaning.

Scrape off any Loctite® residue, if present, from the bore of the valve camshaft gear. Clean the bore in the same manner as the camshaft nose.

Clean all debris from the cylinder head and the camshaft bushings.

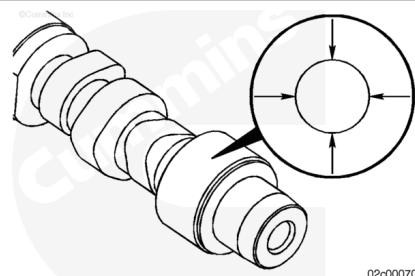
Check the valve camshaft and valve camshaft gear for fretting, pitting, and other damage.

The valve camshaft and valve camshaft gear **must** be replaced if damaged.



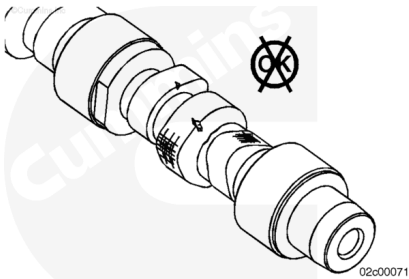
Measure the outside diameter of the seven valve camshaft bearing journals.

Valve Camshaft Bearing Journal Outside Diameter		
mm		in
64.987	MIN	2.5585
65.013	MAX	2.5596



If the camshaft journals are **not** within specifications, the camshaft **must** be replaced.

Inspect the camshaft lobes for cracks, scratches, and other damage. Refer to the Camshaft Reuse Guidelines for Cummins® Engines with Roller Followers or Roller Tappets, Bulletin 3666052 ([/qs3/pubsys2/xml/en/bulletin/3666052.html](http://qs3/pubsys2/xml/en/bulletin/3666052.html)), for further reuse guidelines.



If the valve camshaft and/or the rocker levers need to be replaced, use the Camshaft Compatibility Matrix to determine whether the camshaft and rocker lever roller combination being installed is compatible.

If the matrix states that the combination of the camshaft and rocker lever roller is “No”, the correct combination **must** be installed using the Valve Camshaft Compatibility Matrix.

The wide rocker lever roller is 17 mm [0.67 in] and the narrow rocker lever roller is 15 mm [0.59 in].

The wide camshaft lobe is 20 mm [0.79 in] and the narrow camshaft lobe is 18.5 mm [0.73 in].

Valve Camshaft Compatibility Matrix	Wide Camshaft Lobe	Narrow Camshaft Lobe
Wide Rocker Lever Roller	Yes	No
Narrow Rocker Lever Roller	Yes	Yes

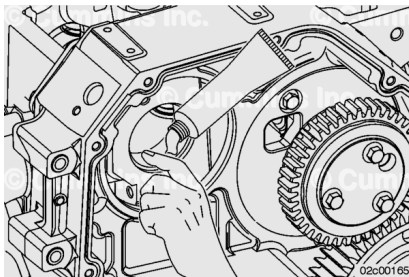
Install

⚠ WARNING ⚠

To reduce the possibility of personal injury, wear protective gloves when handling parts with sharp edges.

⚠ CAUTION ⚠

Use caution when installing the camshaft to avoid damaging the camshaft bushings.



Coat the inside of the valve camshaft bushings with Lubriplate™, Part Number 3163086, or equivalent, prior to installing the camshaft.

Use the valve camshaft installation and removal service tool, Part Number 5299837, or equivalent, to install the camshaft into the cylinder head.

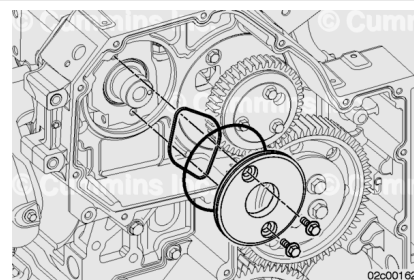
Install the camshaft by hand until three journals are in the cylinder head. Use the Camshaft Installation and Removal Kit, Part Number 5395348, to pull and guide the camshaft into the cylinder head.

Install a new o-ring and seal onto the retainer plate.

Install the retainer plate and capscrews.

Tighten the capscrews.

Torque Value: 40 n•m [30 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the valve camshaft gear and tone wheel. Refer to Procedure 001-088 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-088.html>)
- Install the rocker lever shafts. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/377/377-003-009.html>)
- Connect the internal actuator wiring harness. Refer to Procedure 019-063 in Section 19. (</qs3/pubsys2/xml/en/procedures/377/377-019-063.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/377/377-003-011-tr.html>)
- Install the EGR crossover tube. Refer to Procedure 011-070 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-070-tr.html>)
- Install the upper gear cover. Refer to Procedure 001-079 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-079.html>)
- Install the air compressor inlet tube, if equipped. Refer to Procedure 012-019 in Section 12. (</qs3/pubsys2/xml/en/procedures/377/377-012-019.html>)

- Connect the batteries. Refer to the equipment manufacturer service information.
- Operate the engine to normal operating temperature. Check for leaks and proper operation.

Last Modified: 17-Sep-2019
